

CS5488: BIG DATA ALGORITHMS AND TECHNIQUES

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Big Data Algorithms and Techniques

Subject Code

CS - Computer Science

Course Number

5488

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

CS3402 Database Systems or
CS5481 Data Engineering

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course is aimed at equipping students with the ability to manage very large data sets (Big Data) using a cluster of commodity machines with the main focus on the Hadoop ecosystem. It has three specific objectives: (1) to familiarize students with software systems and techniques for implementing distributed data-parallel programs, (2) to provide insight into internal mechanisms of large-scale data analytical systems, and (3) to acquaint students with big data solutions deployed in real-world settings. Students will also have the opportunity to analyse and to compare real-world big data solutions in a class project case study.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Identify and explain data parallelism to be exploited in large-scale data processing problems.		x	x	
2	Implement data parallel algorithms using techniques covered in the course.		x	x	
3	Describe and explain the internal mechanisms of the Hadoop framework.		x		
4	Design scalable solutions to a real-world problem and sufficiently provide rationalizations to the design decisions.		x	x	x
5	Analyse existing big data solutions deployed in real-world settings through case studies.		x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will learn (1) different types of data-parallel problems; (2) the APIs and tools for handling big data; (3) the internal mechanisms for job scheduling, failure handling, and task execution of the Hadoop framework; (4) case studies on real-world big data algorithms and solutions.	1, 2, 3	2 hours/ week
2	Lab	Students will have the opportunity to (1) familiarize themselves with different big data tools; (2) implement data-parallel algorithms; (3) design experimental studies.	2, 3, 4	1 hour/ week
3	Class Project	Students will work on a use case study on Hadoop-based solutions deployed in real-world settings.	4, 5	

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Class Project	1, 2, 4, 5	40	-	Yes
2	Lab Sheets	1, 2, 3, 4	10	-	Yes

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Group Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

- 1.1 Ability to identify challenges in various types of data parallel problems
- 1.2 Ability to critique existing big data solutions

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Inadequate

Assessment Task

Lab Sheets (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

- 2.1 Ability to contribute to discussions on principle and concepts of scalable data processing

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Inadequate

Assessment Task

Final Exam (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

- 3.1 Ability to demonstrate a good understanding of materials covered in the course

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Inadequate

Assessment Task

Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

1.1 Ability to identify challenges in various types of data parallel problems

1.2 Ability to critique existing big data solutions

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Inadequate

Assessment Task

Lab Sheets (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

2.1 Ability to contribute to discussions on principle and concepts of scalable data processing

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Inadequate

Assessment Task

Final Exam (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

3.1 Ability to demonstrate a good understanding of materials covered in the course

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Inadequate

Part III Other Information

Keyword Syllabus

Big Data, Analytics, MapReduce, Distributed File System, Parallel Processing, Data-parallel Systems, RDBMS, NoSQL, Distributed Indexes, Key-value Stores, Query Languages, Data Manipulation Languages, Consistency, Reliability, Commodity Cluster, Failure Handling, In-memory Processing, Use Case Studies, Emerging Technologies for Big Data Computing (e.g. Hadoop and Spark).

Reading List

Compulsory Readings

Title	
1	Tom White. Hadoop: The Definitive Guide. 4th edition.
2	Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia. Learning Spark: Lightning-Fast Big Data Analysis. 1st edition.

Additional Readings

Title	
1	EMC Education Services. Data Science and Big Data Analytics. 1st edition.